Appendix H - Per-CPU Symbol Index

Entry points, ABI conventions, and reserved memory regions for each CPU. The detailed register descriptions and the per-CPU chapter give the full story; this appendix is the cheat sheet.

H.1 IE64

Symbol	Value / role
Reset vector	`0x000000` (first instruction at start of RAM).
Trap vector base	`0x000400` (8-byte entries, indexed by trap number).
Supervisor stack	grows down from `0x0A0000`.
User stack (`R31`)	grows down from BASIC's per-program stack region.
Call ABI	Arguments R1-R8; result R1; callee-saved R16-R23; caller-saved R1-R15, R24-R30; `R0 = 0` always.
FPU regs	F0-F31, doubles. F0-F7 argument / result, F16-F23 callee-saved.
BASIC text	`0x023000-0x04FFFF` (`BASIC_PROG_START-BASIC_PROG_LIMIT - 1`).
Simple vars	`0x050000-0x057FFF`.
String vars	`0x058000-0x05FFFF`.
Arrays	`0x060000-0x08BFFF`.
String temps	`0x08C000-0x08FFFF`.
GOSUB / FOR stack	`0x090000-0x096FFF`.

H.2 IE32

Symbol	Value / role
Reset vector	`0x000000`.
Stack base	`0x09F000` (`STACK_START`); grows down.
Timer count	MMIO `0xF0804`.
Timer period	MMIO `0xF0808`.
Call ABI	Arguments A,X,Y,Z; result A; B-W caller-saved; stack via PUSH / POP.

H.3 6502

Symbol	Value / role
Reset vector	`$FFFC` (low) / `$FFFD` (high).
IRQ / BRK vector	`$FFFE` / `$FFFF`.
NMI vector	`$FFFA` / `$FFFB`.

Symbol	Value / role
Stack page	`$0100-$01FF`, indexed by `S`.
Zero page	`$0000-$00FF`.
Bank registers	`$F700-$F705`, `$F7F0`.
MMIO aperture	`$F000-$FFF9`, mirrors `0xF0000-0xF0FF9`.
VGA C64-style	`$D700-$D70D`.
ULA paged port	`$D800-$D817`.
PSG / SID	`$D400-$D40F`, `$D500-$D55F`.
POKEY	`$D200-$D209`.
TED audio	`$D600-$D605`.

H.4 Z80

Symbol	Value / role
Reset vector	`0x0000`.
NMI vector	`0x0066`.
RST n	`n * 8` (`0x00`, `0x08`, ..., `0x38`).
IM 2 vector base	`(I << 8) | (data byte from device)`.
Bank registers	port-mapped through bus translation at `$F700-$F705`.
MMIO aperture	`0xF000-0xFFF9`.
PSG / AY ports	`$F0` select, `$F1` data.
TED audio ports	`$F2`, `$F3`.
POKEY ports	`$D0`, `$D1`.
SID ports	`$E0`, `$E1`.
SN76489 ports	`$E4` data, `$E5` status.
VGA ports	`$A0-$AA`.

H.5 M68K 68020

Symbol	Value / role
Reset vector	`$0000.0000` (initial SSP) / `$0000.0004` (initial PC).
Vector table	`$0000.0000-$0000.03FC` (256 entries, 4 bytes each).
Bus error	vector 2.
Address error	vector 3.
Illegal	vector 4.
Zero divide	vector 5.

Symbol	Value / role
CHK	vector 6.
Trapv	vector 7.
Privilege violation	vector 8.
Trace	vector 9 (T1/T0 not enforced in IE; see Ch 28).
Line A	vector 10.
Line F	vector 11.
TRAP #n	vectors 32-47.
Auto-vector IRQs	vectors 25-31.
Call ABI	Arguments on stack; D0 / A0 caller-saved; D2-D7 / A2-A6 callee-saved.

H.6 x86 (8086 + 386 extensions, real-mode only)

Symbol	Value / role
Reset vector	`EIP = 0`, `CS = 0`, `DS = ES = SS = 0` (flat, not the 8086 `F000:FFF0` boot vector).
IVT	`0x0000`-`0x03FF` (256 entries, 4 bytes each: offset + segment).
Stack	`SS:ESP`, segments are zero so the stack lives in flat RAM.
Call ABI	Caller pushes arguments right-to-left; `EAX` returns; `EBX`, `ESI`, `EDI`, `EBP` callee-saved; `ECX`, `EDX` caller-saved.
BIOS-style ints	reserved; no BIOS ROM is provided. The IVT is initialised to a default IRET routine.

Real-mode 20-bit physical address calculation `(seg << 4) + ofs` is performed in software for compatibility; the 32-bit linear form (the result of the calculation) is what reaches the bus. Programs that use 32-bit immediate addressing reach the full flat address space directly.

H.7 Cross-CPU bus addresses (shared)

These addresses are the same in every CPU's 32-bit view of the bus. The 8-bit CPUs reach them through the bank-window mechanism described in Chapters 26 and 27.

Address	Meaning
`0xF0700`	`TERM_OUT`.
`0xF1400`	HOST appliance block.
`0xF2200`	File I/O block.
`0xF2300`	Media loader.
`0xF2320`	Image executor.
`0xF2340`	Coprocessor.
`0xF2400`	SysInfo.
`0xF8000`	Voodoo 3D.